

Table 1. Biological context correspondence between Tox21 assays and LINCS transcriptomic profiles. Tox21 endpoints were matched to available LINCS profiles according to cellular correspondence, exposure time, and mechanistic relevance. Cellular correspondence was classified as exact, tissue-of-origin proxy, or unmatched. The rightmost column sums cellular (0–2), temporal (0–1), and mechanistic (0–1) correspondence; each star is one point, and a dash in the score column denotes zero. This qualitative score was assigned before model evaluation and is not an empirical performance measure.

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Table 2. Molecular-model performance across the twelve Tox21 endpoints. Endpoint-specific AUPRC and macro-AUPRC are reported for the nuclear-receptor (NR) and stress-response (SR) groups. The best and second-best values in each column are shown in bold and underlined, respectively. The random-baseline AUPRC equals endpoint activity prevalence in the molecular evaluation set. Molecular predictors are frozen; their performance is unchanged by the revised transcriptomic fitting procedure.

Model	NR-AR	NR-AR-LBD	NR-AhR	NR-Aromat	NR-ER	NR-ER-LBD	NR-PPAR-g	NR macro	SR-ARE	SR-ATAD5	SR-HSE	SR-MMP	SR-p53	SR macro
GAT	0.138	0.087	0.365	0.165	0.263	0.049	0.073	0.163	0.371	0.140	0.129	0.351	0.211	0.240
MACCS + logistic regression	0.696	0.273	0.319	0.206	0.370	0.130	0.057	0.293	0.377	0.086	0.154	0.473	0.292	0.276
RDKit descriptors + XGBoost	0.616	0.341	0.472	0.221	0.307	0.195	0.150	0.329	0.480	0.178	0.154	0.475	0.292	0.316
RDKit descriptors + logistic regression	0.706	0.339	0.445	0.251	0.238	0.194	0.193	0.338	0.470	0.177	0.144	0.552	0.286	0.326
MACCS + XGBoost	0.616	0.374	0.375	0.360	0.384	0.264	0.101	0.353	0.539	0.110	0.111	0.513	0.339	0.323
Morgan + XGBoost	0.764	0.381	0.424	0.370	0.331	0.153	0.246	0.381	0.428	0.159	0.157	0.474	0.247	0.293
Morgan + logistic	0.775	0.231	0.491	0.252	0.309	0.150	0.234	0.349	0.450	0.205	0.226	0.537	0.327	0.349
ChemBERTa	0.744	0.367	0.450	0.283	0.415	0.161	0.044	0.352	0.504	0.180	0.203	0.562	0.301	0.350
Chemprop	0.734	0.232	0.373	0.251	0.335	0.121	0.308	0.336	0.432	0.188	0.313	0.533	0.397	0.373
GCN	0.736	0.306	0.432	0.237	0.388	0.167	0.103	0.339	0.500	0.116	0.383	0.572	0.390	0.392
GraphSAGE	0.802	0.342	0.448	0.350	0.376	0.183	0.074	0.368	0.508	0.131	0.321	0.544	0.431	0.387
GIN	0.818	0.370	0.512	0.309	0.386	0.166	0.072	0.376	0.569	0.212	0.456	0.525	0.436	0.440
Task-specific K=4 arithmetic mean	0.821	0.415	0.574	0.444	0.460	0.219	0.322	0.465	0.618	0.341	0.432	0.622	0.527	0.508
Random baseline	0.038	0.040	0.159	0.074	0.140	0.046	0.045	0.077	0.279	0.058	0.089	0.200	0.136	0.152

Table S1. Effect of cohort-level intercept adjustment on probabilistic fit and discrimination in selected transcriptomic residual models. Results are shown for NR-Aromatase, SR-ARE, SR-MMP, and SR-p53 across the evaluated cell lines in the primary 6-h cohorts. Both $\Delta\log$ loss and ΔAUPRC compare transcriptomic models fitted with calibrated and uncalibrated frozen molecular offsets (D1 – C1). In D1, the intercept is fitted first and held fixed during β estimation. Both biological branches use unadjusted expression with training-fold standardization, unweighted BCE, and independently selected ridge penalties. Negative $\Delta\log$ loss indicates improved probabilistic fit, whereas positive ΔAUPRC indicates improved discrimination. Estimates are means $\pm$ SD across 20 repeat-level means, each averaging five outer folds; 95% confidence intervals and two-sided nominal P values use the Nadeau-Bengio correction over 100 paired outer-fold effects.

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Table S2. Context-specific predictive performance of chemical and transcriptome-complemented models across the twelve Tox21 endpoints. AUPRC is reported to three decimal places for Chem. and Chem. + Bio. on matching held-out compounds within each endpoint, cell line, and exposure time. The 6-h rows represent primary full cohorts; the 24-h rows represent paired exposure-time cohorts. The frozen chemical logit is calibrated on each training partition; this intercept is held fixed while transcriptomic β coefficients are fitted using unweighted BCE plus a ridge penalty selected by inner cross-validation. Unadjusted perturbation expression is standardized using training-fold means and standard deviations. Bold Chem. + Bio. values and asterisks denote nominally significant positive improvements over the corresponding Chem. value (*P < 0.05, two-sided Nadeau-Bengio corrected resampled test, without multiple-testing adjustment). Endpoint labels are bold when at least one displayed context meets this criterion. † denotes fewer than 20 active compounds. Paired-cohort 6-h results are retained in the full numeric source but are not displayed.

Endpoint	Cell	Time (h)	n	Active	Chem.	Chem. + Bio.
NR-AR	HA1E	6	968	72	0.461	0.461
NR-AR	HepG2	6	668	46	0.537	0.537
NR-AR	HT29	6	960	72	0.457	0.457
NR-AR	MCF7	6	1259	82	0.459	0.459
NR-AR	HA1E	24	830	59	0.432	0.432
NR-AR	HT29	24	585	44	0.385	0.385
NR-AR	MCF7	24	1243	78	0.460	0.460
NR-AR-LBD	HA1E	6	869	49	0.554	0.554
NR-AR-LBD	HepG2	6	598	37	0.554	0.554
NR-AR-LBD	HT29	6	862	47	0.561	0.561
NR-AR-LBD	MCF7	6	1145	53	0.581	0.585
NR-AR-LBD	HA1E	24	747	39	0.503	0.503
NR-AR-LBD	HT29	24	523	19	0.510	0.510†
NR-AR-LBD	MCF7	24	1129	51	0.583	0.586
NR-AhR	HA1E	6	849	102	0.526	0.526
NR-AhR	HepG2	6	585	65	0.477	0.482
NR-AhR	HT29	6	841	95	0.479	0.479
NR-AhR	MCF7	6	1111	112	0.473	0.472
NR-AhR	HA1E	24	727	91	0.527	0.530
NR-AhR	HT29	24	516	63	0.521	0.521
NR-AhR	MCF7	24	1099	111	0.468	0.468
NR-Aromatase	HA1E	6	759	63	0.384	0.384
NR-Aromatase	HepG2	6	512	47	0.346	0.356
NR-Aromatase	HT29	6	758	63	0.372	0.378
NR-Aromatase	MCF7	6	1021	76	0.409	0.431
NR-Aromatase	HA1E	24	655	57	0.397	0.482
NR-Aromatase	HT29	24	475	45	0.371	0.371
NR-Aromatase	MCF7	24	1012	76	0.411	0.538*
NR-ER	HA1E	6	805	132	0.430	0.430
NR-ER	HepG2	6	540	92	0.477	0.477
NR-ER	HT29	6	799	128	0.419	0.419
NR-ER	MCF7	6	1067	155	0.402	0.402
NR-ER	HA1E	24	689	112	0.441	0.441
NR-ER	HT29	24	487	76	0.409	0.409
NR-ER	MCF7	24	1054	154	0.407	0.407
NR-ER-LBD	HA1E	6	907	55	0.385	0.384
NR-ER-LBD	HepG2	6	620	39	0.404	0.404
NR-ER-LBD	HT29	6	902	53	0.397	0.398
NR-ER-LBD	MCF7	6	1194	61	0.379	0.418
NR-ER-LBD	HA1E	24	772	40	0.411	0.400
NR-ER-LBD	HT29	24	547	25	0.375	0.369
NR-ER-LBD	MCF7	24	1178	59	0.397	0.429
NR-PPAR-gamma	HA1E	6	811	34	0.235	0.235
NR-PPAR-gamma	HepG2	6	552	19	0.300	0.299†
NR-PPAR-gamma	HT29	6	803	29	0.193	0.192
NR-PPAR-gamma	MCF7	6	1073	34	0.222	0.222
NR-PPAR-gamma	HA1E	24	696	29	0.241	0.241
NR-PPAR-gamma	HT29	24	483	17	0.278	0.277†

Table S2 continued

Endpoint	Cell	Time (h)	n	Active	Chem.	Chem. + Bio.
NR-PPAR-gamma	MCF7	24	1060	33	0.212	0.212
SR-ARE	HA1E	6	690	136	0.474	0.479
SR-ARE	HepG2	6	451	98	0.452	0.553*
SR-ARE	HT29	6	682	129	0.451	0.452
SR-ARE	MCF7	6	923	160	0.452	0.459
SR-ARE	HA1E	24	584	118	0.469	0.482
SR-ARE	HT29	24	416	80	0.498	0.492
SR-ARE	MCF7	24	913	159	0.449	0.463
SR-ATAD5	HA1E	6	911	50	0.295	0.293
SR-ATAD5	HepG2	6	619	33	0.372	0.368
SR-ATAD5	HT29	6	903	50	0.307	0.316
SR-ATAD5	MCF7	6	1196	57	0.254	0.260
SR-ATAD5	HA1E	24	778	43	0.311	0.337
SR-ATAD5	HT29	24	544	30	0.308	0.308
SR-ATAD5	MCF7	24	1180	55	0.257	0.319
SR-HSE	HA1E	6	814	52	0.234	0.248
SR-HSE	HepG2	6	547	30	0.269	0.252
SR-HSE	HT29	6	806	50	0.254	0.251
SR-HSE	MCF7	6	1077	64	0.248	0.245
SR-HSE	HA1E	24	697	43	0.231	0.238
SR-HSE	HT29	24	498	31	0.297	0.299
SR-HSE	MCF7	24	1065	62	0.239	0.254
SR-MMP	HA1E	6	723	129	0.588	0.646*
SR-MMP	HepG2	6	495	102	0.616	0.686*
SR-MMP	HT29	6	717	121	0.563	0.643*
SR-MMP	MCF7	6	958	143	0.527	0.635*
SR-MMP	HA1E	24	613	111	0.571	0.681*
SR-MMP	HT29	24	430	61	0.547	0.553
SR-MMP	MCF7	24	947	141	0.529	0.659*
SR-p53	HA1E	6	884	89	0.361	0.404
SR-p53	HepG2	6	603	64	0.344	0.395
SR-p53	HT29	6	877	85	0.338	0.427
SR-p53	MCF7	6	1158	100	0.295	0.397*
SR-p53	HA1E	24	757	76	0.350	0.381
SR-p53	HT29	24	533	57	0.374	0.399
SR-p53	MCF7	24	1142	98	0.294	0.508*

Table S3. Classical molecular-model settings. Each fingerprint was used in a separate endpoint-specific model. Shared settings are listed once; braces indicate hyperparameter search spaces. Model selection used validation AUPRC. These frozen molecular-model settings are unchanged by the revision to transcriptomic fitting. Shared settings: training seed: 1.

Representation	Size	Model	Selection / tune	Class weighting	Settings
MACCS structural	167.0	Logistic regression	C in {0.001, 0.01, 0.1, 1, 10, 100}; selected by validation AUPRC	class_weight=balanced	max_iter=2000
Morgan fingerprint	2048.0	Logistic regression	radius=2; C in {0.001, 0.01, 0.1, 1, 10, 100}; selected by validation AUPRC	class_weight=balanced	max_iter=2000
RDKit topological fingerprint	2048.0	Logistic regression	C in {0.001, 0.01, 0.1, 1, 10, 100}; selected by validation AUPRC	class_weight=balanced	max_iter=2000
MACCS structural	167.0	XGBoost	max_depth in {3, 4, 6}; learning_rate in {0.05, 0.1}; selected by validation AUPRC	scale_pos_weight=1	n_estimators=300; subsample=0.8; colsample_bytree=0.8; early_stopping_rounds=10; eval_metric=auprc; n_jobs=1
Morgan fingerprint	2048.0	XGBoost	radius=2; max_depth in {3, 4, 6}; learning_rate in {0.05, 0.1}; selected by validation AUPRC	scale_pos_weight=1	n_estimators=300; subsample=0.8; colsample_bytree=0.8; early_stopping_rounds=10; eval_metric=auprc; n_jobs=1
RDKit topological fingerprint	2048.0	XGBoost	max_depth in {3, 4, 6}; learning_rate in {0.05, 0.1}; selected by validation AUPRC	scale_pos_weight=1	n_estimators=300; subsample=0.8; colsample_bytree=0.8; early_stopping_rounds=10; eval_metric=auprc; n_jobs=1

Table S4. Deep molecular-model settings. All models used batches of 32. GNN denotes the four pretrained architectures: GCN, GAT, GIN, and GraphSAGE. These frozen molecular-model settings, including their original class weighting, are unchanged; unweighted BCE applies to the revised transcriptomic stage. Shared settings: training seed: 1.

Representation	Size	Model	Selection / tune	Class weighting	Settings
Molecular graph	300.0	GCN	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Molecular graph	300.0	GAT	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Molecular graph	300.0	GIN	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Molecular graph	300.0	GraphSAGE	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Canonical SMILES tokens		DeepChem/Chem	End-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	max_length=128; batch_size=32; Adam; lr=1e-4; weight_decay=0; linear warmup=6; gradient clipping=1.0; max_epochs=20; patience=5
Directed molecular graph	300.0	Chemprop D-MPN	Single-task training; early stopping on validation loss	Unweighted binary classification loss	hidden_dim=300; depth=3; mean aggregation; dropout=0; batch_size=32; max_epochs=100; patience=20

Table S5. Transcriptomic residual-model and validation settings. The molecular predictor remains frozen with coefficient one. The cohort intercept is fitted first using unweighted BCE on each training partition and held fixed while transcriptomic β coefficients are estimated. β is fitted using unweighted BCE plus an L2 penalty; its strength is selected independently within each outer-training partition by three-fold inner cross-validation using unpenalized mean BCE and the largest penalty within one standard error of the global minimum. Unadjusted perturbation expression receives training-fold standardization only. In executable code, alpha denotes the ridge penalty λ , not the calibration intercept. The outer and inner procedures are specified in this table.

Section	Setting	Value
Study design	Chemical prior	Endpoint-specific top-4 arithmetic mean
Study design	Chemical prior coeffic	Fixed at 1.0
Study design	Stage-2 intercept	Fit an unpenalized offset-only intercept on each inner/outer training partition for β estimation and predictions
Study design	Transcriptomic	β only; minimize mean unweighted BCE + $(\lambda/2) \beta ^2$ with the calibrated molecular predictor
Input	Cell lines	HA1E, HEPG2, HT29, MCF7
Input	Primary exposure	Primary full-cohort 6 h; paired 6-h and 24-h cohorts evaluated separately
Input	Dose window	8-12 uM
Input	Genes	978 L1000 landmark genes
Input	Replicate aggregation	Feature-wise median
Preprocessing	No-adjustment variati	Unadjusted perturbation expression; feature-wise training mean/SD (ddof=1e-8 by 1; apply training scaler to validation/test)
Validation	Outer repeats	20
Validation	Outer folds per repeat	5
Validation	Outer grouping	Bemis-Murcko scaffold
Validation	Distinct outer partition	Required; duplicate signatures forbidden
Validation	Minimum class count	At least 2 positives and 2 negatives in outer train and test
Validation	Inner folds	3
Validation	Inner grouping	Stratified Bemis-Murcko scaffold groups
Validation	Inner selection metric	Unpenalized mean BCE computed stably from logits on inner-validation fold
Validation	Ridge selection rule	Select the largest λ anywhere on the grid within one SE of the global minimum inner-validation BCE; SE is sample SD across three folds / sqrt(3)
Optimization	Ridge alpha grid	$\lambda = 10^{(-4 + k/2)}$, k=0,...,24 (25 values); called alpha in the executable; sorted in descending order
Optimization	Solver	L-BFGS-B with analytical gradient
Optimization	Primary max iteration	500
Optimization	Cold-retry max iteration	2000
Optimization	Gradient tolerance	1e-6
Optimization	Acceptance gradient	1e-5
Optimization	Function tolerance	1e-14
Likelihood	Primary objective	Mean unweighted binary cross-entropy (equal compound weights); no class weights
Metric	Primary discrimination metric	AUPRC
Metric	Secondary discrimination metric	AUROC
Metric	Calibration metrics	Log loss and Brier score
Inference	Paired effect	Chem. + Bio. minus intercept-calibrated Chem. on identical held-out compounds
Inference	Dependence correction	Nadeau-Bengio corrected resampled paired t-test on 100 outer-fold effects
Inference	Multiple testing	Benjamini-Hochberg within variant and cell across 12 endpoints
Interpretation	Confirmatory active-compound threshold	At least 20 active compounds per endpoint-cell cohort
Reproducibility	Stage-1 seed	1
Reproducibility	Completed unadjusted fits	Primary 6 h: 4,800; paired 6 h: 3,600; paired 24 h: 3,600. All requested fits completed with zero failures. These counts exclude the separate residualized sensitivity validation

Table S6. Nuclear-receptor cohort sizes and analysis eligibility. n is the number of labeled compounds; active counts and percentages refer to these compounds. Symbols denote primary (✓), exploratory (△), and excluded (x) analyses; bold active counts are below the primary-analysis threshold of 20. Total profiled compounds (6 h/paired 6–24 h) are HA1E: 1,012/861; HepG2: 695/46; HT29: 1,005/606; MCF7: 1,332/1,316. Missing-label and inactive counts can be obtained by subtraction. Cohort membership is unchanged by the estimator revision; fold-size ranges in the numeric source are checked against the corrected runs.

Tox21 endpoint	Cell line	Exposure	n	Active	Active (%)	Symbol
NR-AR	HA1E	6	968	72	7.4	✓
NR-AR	HepG2	6	668	46	6.9	✓
NR-AR	HT29	6	960	72	7.5	✓
NR-AR	MCF7	6	1259	82	6.5	✓
NR-AR-LBD	HA1E	6	869	49	5.6	✓
NR-AR-LBD	HepG2	6	598	37	6.2	✓
NR-AR-LBD	HT29	6	862	47	5.5	✓
NR-AR-LBD	MCF7	6	1145	53	4.6	✓
NR-AhR	HA1E	6	849	102	12.0	✓
NR-AhR	HepG2	6	585	65	11.1	✓
NR-AhR	HT29	6	841	95	11.3	✓
NR-AhR	MCF7	6	1111	112	10.1	✓
NR-Aromatase	HA1E	6	759	63	8.3	✓
NR-Aromatase	HepG2	6	512	47	9.2	✓
NR-Aromatase	HT29	6	758	63	8.3	✓
NR-Aromatase	MCF7	6	1021	76	7.4	✓
NR-ER	HA1E	6	805	132	16.4	✓
NR-ER	HepG2	6	540	92	17.0	✓
NR-ER	HT29	6	799	128	16.0	✓
NR-ER	MCF7	6	1067	155	14.5	✓
NR-ER-LBD	HA1E	6	907	55	6.1	✓
NR-ER-LBD	HepG2	6	620	39	6.3	✓
NR-ER-LBD	HT29	6	902	53	5.9	✓
NR-ER-LBD	MCF7	6	1194	61	5.1	✓
NR-PPAR-γ	HA1E	6	811	34	4.2	✓
NR-PPAR-γ	HepG2	6	552	19	3.4	△
NR-PPAR-γ	HT29	6	803	29	3.6	✓
NR-PPAR-γ	MCF7	6	1073	34	3.2	✓
NR-AR	HA1E	6 and 24	830	59	7.1	✓
NR-AR	HepG2	6 and 24	45	5	11.1	x
NR-AR	HT29	6 and 24	585	44	7.5	✓
NR-AR	MCF7	6 and 24	1243	78	6.3	✓
NR-AR-LBD	HA1E	6 and 24	747	39	5.2	✓
NR-AR-LBD	HepG2	6 and 24	31	1	3.2	x
NR-AR-LBD	HT29	6 and 24	523	19	3.6	△
NR-AR-LBD	MCF7	6 and 24	1129	51	4.5	✓
NR-AhR	HA1E	6 and 24	727	91	12.5	✓
NR-AhR	HepG2	6 and 24	36	6	16.7	x
NR-AhR	HT29	6 and 24	516	63	12.2	✓
NR-AhR	MCF7	6 and 24	1099	111	10.1	✓
NR-Aromatase	HA1E	6 and 24	655	57	8.7	✓
NR-Aromatase	HepG2	6 and 24	26	14	53.8	x
NR-Aromatase	HT29	6 and 24	475	45	9.5	✓
NR-Aromatase	MCF7	6 and 24	1012	76	7.5	✓
NR-ER	HA1E	6 and 24	689	112	16.3	✓
NR-ER	HepG2	6 and 24	23	11	47.8	x
NR-ER	HT29	6 and 24	487	76	15.6	✓
NR-ER	MCF7	6 and 24	1054	154	14.6	✓
NR-ER-LBD	HA1E	6 and 24	772	40	5.2	✓
NR-ER-LBD	HepG2	6 and 24	39	8	20.5	x
NR-ER-LBD	HT29	6 and 24	547	25	4.6	✓
NR-ER-LBD	MCF7	6 and 24	1178	59	5.0	✓
NR-PPAR-γ	HA1E	6 and 24	696	29	4.2	✓

Table S6 continued

Tox21 endpoint	Cell line	Exposure	n	Active	Active (%)	Symbol
NR-PPAR-γ	HepG2	6 and 24	29	2	6.9	×
NR-PPAR-γ	HT29	6 and 24	483	17	3.5	△
NR-PPAR-γ	MCF7	6 and 24	1060	33	3.1	✓

Table S7. Stress-response cohort sizes and analysis eligibility. n is the number of labeled compounds; active counts and percentages refer to these compounds. Symbols denote primary (✓), exploratory (△), and excluded (×) analyses; bold active counts are below the primary-analysis threshold of 20. Total profiled compounds (6 h/paired 6–24 h) are HA1E: 1,012/861; HepG2: 695/46; HT29: 1,005/606; MCF7: 1,332/1,316. Missing-label and inactive counts can be obtained by subtraction. Cohort membership is unchanged by the estimator revision; fold-size ranges in the numeric source are checked against the corrected runs.

Tox21 endpoint	Cell line	Exposure	n	Active	Active (%)	Symbol
SR-ARE	HA1E	6	690	136	19.7	✓
SR-ARE	HepG2	6	451	98	21.7	✓
SR-ARE	HT29	6	682	129	18.9	✓
SR-ARE	MCF7	6	923	160	17.3	✓
SR-ATAD5	HA1E	6	911	50	5.5	✓
SR-ATAD5	HepG2	6	619	33	5.3	✓
SR-ATAD5	HT29	6	903	50	5.5	✓
SR-ATAD5	MCF7	6	1196	57	4.8	✓
SR-HSE	HA1E	6	814	52	6.4	✓
SR-HSE	HepG2	6	547	30	5.5	✓
SR-HSE	HT29	6	806	50	6.2	✓
SR-HSE	MCF7	6	1077	64	5.9	✓
SR-MMP	HA1E	6	723	129	17.8	✓
SR-MMP	HepG2	6	495	102	20.6	✓
SR-MMP	HT29	6	717	121	16.9	✓
SR-MMP	MCF7	6	958	143	14.9	✓
SR-p53	HA1E	6	884	89	10.1	✓
SR-p53	HepG2	6	603	64	10.6	✓
SR-p53	HT29	6	877	85	9.7	✓
SR-p53	MCF7	6	1158	100	8.6	✓
SR-ARE	HA1E	6 and 24	584	118	20.2	✓
SR-ARE	HepG2	6 and 24	26	15	57.7	×
SR-ARE	HT29	6 and 24	416	80	19.2	✓
SR-ARE	MCF7	6 and 24	913	159	17.4	✓
SR-ATAD5	HA1E	6 and 24	778	43	5.5	✓
SR-ATAD5	HepG2	6 and 24	33	6	18.2	×
SR-ATAD5	HT29	6 and 24	544	30	5.5	✓
SR-ATAD5	MCF7	6 and 24	1180	55	4.7	✓
SR-HSE	HA1E	6 and 24	697	43	6.2	✓
SR-HSE	HepG2	6 and 24	36	6	16.7	×
SR-HSE	HT29	6 and 24	498	31	6.2	✓
SR-HSE	MCF7	6 and 24	1065	62	5.8	✓
SR-MMP	HA1E	6 and 24	613	111	18.1	✓
SR-MMP	HepG2	6 and 24	29	19	65.5	×
SR-MMP	HT29	6 and 24	430	61	14.2	✓
SR-MMP	MCF7	6 and 24	947	141	14.9	✓
SR-p53	HA1E	6 and 24	757	76	10.0	✓
SR-p53	HepG2	6 and 24	38	15	39.5	×
SR-p53	HT29	6 and 24	533	57	10.7	✓
SR-p53	MCF7	6 and 24	1142	98	8.6	✓